

Tarun Sharma

Engineer-Founder · **Software-Defined Vehicles** · Full-Stack & Cloud Infrastructure

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Hands-on engineer-founder who ships across the stack — from embedded Linux and open-source infrastructure to AI-native products. Spent **5 years architecting software-defined vehicle platforms at Jaguar Land Rover**, presenting base-OS progress to the JLR and Tata Motors boards; now building a **Digital Twin venture** and shipping full-stack AI products end-to-end. I work where mechanical engineering, software, and AI intersect.

// EXPERIENCE

Twinity Labs Founder & CEO

Sep 2025 – Present
Bengaluru, India

- ◆ Founding an applied **Digital Twin** venture; shipping AI-native products end-to-end — Flutter clients, AWS serverless backends (**Lambda, DynamoDB, Cognito**), and **OpenTofu** IaC.
- ◆ Built **SetMind**, an AI strength-training app: offline-first Flutter + Drift/SQLite with bidirectional cloud sync (**16-table** schema), Claude/OpenAI workout generation, and RevenueCat monetization.
- ◆ Drive the full development lifecycle with agentic AI tooling — Claude Code, Cursor, vLLM, LiteLLM.

Jaguar Land Rover

Sep 2020 – Aug 2025 · 5 yrs
Bengaluru, India

Software Engineer → Senior Software Engineer, Software-Defined Vehicles

- ◆ Triggered an enterprise-wide migration from commercial **Terraform to OpenTofu** and an open-source build stack (BuildBarn, Lorry, Trustable) — cutting licensing spend and hardening supply-chain security.
- ◆ Orchestrated **zero-downtime migration of 450+ GitLab repos** and Podman images into JLR, keeping Central Compute Module builds green throughout cutover.
- ◆ Led **OS/BSP bring-up** for a safety-qualified **Linux** platform (kernel, systemd, NXP S32N silicon); drove **ASIL-B / ISO 26262** certification plans and presented to the JLR & Tata Motors boards.
- ◆ Built an automated **AWS boot-observability pipeline** (S3, Lambda, EC2, Redshift → Tableau) that flashes physical SoCs for data-driven regression tracking.
- ◆ Architected a Python **Test Executable Generator** (factory/builder) automating execution across dSPACE / Vector / NI on **300+** test assets; shipped "Mars Attacks," a FastAPI microservice.
- ◆ Filed a **patent (pending)** for LSTM-based predictive maintenance; mentored 4 interns (Rust vs C++, led to PPOs) and 2 graduate engineers. **1st Runner-Up**, JLR global GDD Hackathon.

MKS Enterprises

Jan 2026 – Present
Jamshedpur, India

General Manager // CONCURRENT

- ◆ Lead operations and P&L for a heavy-crane trading and services business serving major industrial and public-sector clients.

// EDUCATION

Indian Institute of Technology Kanpur

2015 – 2020
Kanpur, India

B.Tech–M.Tech Dual Degree, Mechanical Engineering

- ◆ Master's thesis on railway bogie suspension redesign **adopted by Indian Railways** for production rollout across FIAT-type LHB coaches.
- ◆ President, Association of Mechanical Engineers; Teaching Assistant for two courses (200+ students).

// TECHNICAL SKILLS

LANGUAGES	Python (FastAPI), Rust, C++, TypeScript, MATLAB
INFRA & CLOUD	OpenTofu / Terraform, GitLab CI/CD, AWS (Lambda, DynamoDB, S3, EC2, Redshift, Cognito), GCP, Podman, BuildStream, CMake / Meson / Ninja
SYSTEMS	Linux kernel, systemd, embedded bring-up (NXP S32N), RTI DDS, ISO 26262 / ASIL-B, deterministic builds, SBOM / supply-chain security
AI & PRODUCT	Claude & OpenAI APIs, agentic AI tooling, LSTM / CNNs, Flutter, Riverpod, offline-first sync

// RECOGNITION & PUBLICATION

- ◆ **Publication** — "Primary suspension failure analysis in FIAT-type LHB bogies," *Engineering Failure Analysis*, vol. 138 (2022).
- ◆ **1st Runner-Up**, JLR global GDD Hackathon (2021) — gaze-controlled HUDs, across all 6 GDD centers.
- ◆ **Patent (pending)** — LSTM-based predictive maintenance for in-vehicle systems (JLR).
- ◆ **Asha Khanna Award** — national topper in Mathematics, ISC 2015 · JEE Advanced qualified.